Answer Key — Compounds

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1	**(b) combine chemically in a fixed ratio** — a compound = elements chemically combined in a fixed proportion.
Q2	**(b) hydrogen and oxygen** — electrolysis splits water into hydrogen and oxygen.
Q3	**(b) hydrogen** — hydrogen gives the pop sound; oxygen makes the flame brighter.
Q4	**(b) it has new properties and a fixed composition** — compounds have new properties and a fixed composition.
Q5	**(b) False** — compounds can be split only by chemical methods.
Q6	**oxygen** — oxygen supports burning, making the flame brighter.
Q7	**fixed** — a compound has a fixed proportion of elements.

Short answer

Q8	A pure substance formed when two or more elements combine chemically in a fixed ratio, e.g. water (hydrogen + oxygen).
Q9	Mixture: components just mixed, any ratio, keep their properties, separated physically. Compound: elements chemically combined, fixed ratio, new properties, separated only chemically (any three).
Q10	Bring a burning candle near each: a pop sound shows hydrogen; a brighter flame shows oxygen.

Long / application

Q11	In the mixture, iron and sulfur keep their own properties — a magnet can pull out the iron, and the ratio can vary. After heating, they combine chemically into iron sulfide, a compound with new properties — the magnet no longer separates the iron, and the elements are in a fixed proportion. This shows mixtures are not combined while compounds are.
Q12	Passing electricity through slightly acidified water produces gas at both terminals: about twice as much hydrogen as oxygen. The pop test shows hydrogen and the brighter-flame test shows oxygen. This proves water is a compound made of the elements hydrogen and oxygen (so it is not an element).